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Master of Computer Applications

Semester – III

Data Analytics for Business

Experiment No.: 5

Title: Building Sales Performance Dashboards

Submitted By:

Name: Attem Deepak Mudiraj

Reg. No.: 2511022250001

Date: 05/08/2026

Submitted To:

Faculty Name: Mr. Aman Kumar Sharma

Department of Computer Application
Alliance University
Chandapura - Anekal Main Road, Anekal
Bengaluru – 562106

Experiment No.: 5

Building Sales Performance Dashboards

Problem Statement

Develop an interactive Power BI dashboard using the Walmart Weekly Sales Dataset to analyze weekly sales performance across different stores. The dashboard should provide insights into sales trends, store performance, holiday sales, and external factors such as temperature by using interactive visualizations including Bar Charts, Line Charts, Donut Charts, KPI Cards, and Scatter Charts. The dashboard should help management monitor business performance and support data-driven decision-making.

Methodology / Steps Followed

The following methodology was followed to develop the **Walmart Weekly Sales Dashboard** in Power BI:

Step 1: Dataset Collection

- Downloaded the **Walmart Weekly Sales Dataset** from Kaggle.
- Imported the dataset into **Power BI Desktop** for analysis.

Step 2: Data Loading and Preparation

- Loaded the dataset into Power BI using the Get Data option.
- Verified the data types for all columns.
- Converted the Date column to the Date format.
- Checked for missing values and ensured numerical columns were correctly formatted.

Step 3: Data Modeling

- Created DAX measures to calculate key performance indicators (KPIs), including:
 - Total Sales
 - Average Weekly Sales
 - Total Stores
 - Holiday Weeks
- Used these measures to provide summarized business information.

Step 4: Dashboard Layout Design

- Designed a clean and professional dashboard layout.
- Added a dashboard title and organized the report into sections.
- Created a dedicated filter panel on the left side for easy navigation.
- Applied a consistent blue-and-white color theme to improve readability.

Step 5: Interactive Filters

- Added slicers for:
 - Date
 - Store
 - Holiday Flag
 - Temperature
 - Fuel Price
 - CPI
 - Unemployment
- Configured the slicers to interact with all visualizations on the dashboard.

Step 6: KPI Card Creation

- Created KPI cards to display:
 - Total Sales
 - Average Weekly Sales
 - Total Stores
 - Holiday Weeks
- Formatted the cards with appropriate labels for quick business insights.

Step 7: Chart Creation

Created different business charts to analyze sales performance:

- **Line Chart** to visualize weekly sales trends over time.
- **Clustered Bar Chart** to identify the top-performing Walmart stores.
- **Donut Chart** to compare holiday and non-holiday sales.
- **Scatter Chart** to analyze the relationship between temperature and weekly sales.

Step 8: Dashboard Interactivity

- Added navigation buttons for different dashboard sections.
- Created a **Reset Filters** button using bookmarks.
- Enabled cross-filtering so that selecting any visual updates the remaining charts automatically.

Step 9: Dashboard Formatting

- Applied consistent formatting across all visuals.
- Added titles, borders, shadows, and aligned the visuals properly.
- Used appropriate colors and labels to improve the overall appearance of the dashboard.

Step 10: Business Insight Generation

- Analyzed store-wise sales performance.
- Identified weekly sales trends.
- Compared holiday and non-holiday sales.
- Evaluated the influence of temperature on sales.
- Generated business insights to support management decision-making.

Output :

New Measures using “DAX”:

Table: Walmart (6,435 rows) Column: Measure (0 distinct values)

Store	Date	Weekly_Sales	Holiday_Flag	Temperature	Fuel_Price	CPI	Unemployment
2	02-07-2010	2003940.64	0	82.74	2.669	210.8803726	8.099
2	09-07-2010	1880902.62	0	82.59	2.642	210.7657317	8.099
2	16-07-2010	1845879.79	0	85.32	2.623	210.7577954	8.099
2	23-07-2010	1781717.71	0	87.66	2.608	210.8921319	8.099
2	30-07-2010	1804246.16	0	83.49	2.64	211.0264684	8.099
2	06-08-2010	1991909.98	0	89.53	2.627	211.1608049	8.099
2	13-08-2010	1895601.05	0	89.05	2.692	211.2951413	8.099
2	20-08-2010	1964335.23	0	88.7	2.664	211.2596586	8.099
2	27-08-2010	1863840.49	0	87.12	2.619	211.2241759	8.099
2	03-09-2010	1904608.09	0	81.83	2.577	211.1886931	8.099
2	17-09-2010	1793903.6	0	82.05	2.582	211.1806415	8.099
2	24-09-2010	1724557.22	0	81.79	2.624	211.2552578	8.099
15	02-07-2010	70937.11	0	66.13	2.958	132.5218667	8.099
15	09-07-2010	691497.62	0	77.33	2.94	132.5461333	8.099
15	16-07-2010	633203.69	0	72.83	2.933	132.5667742	8.099
15	23-07-2010	598553.43	0	73.2	2.924	132.5825806	8.099
15	30-07-2010	619224.06	0	72.04	2.932	132.5983871	8.099
15	06-08-2010	639651.24	0	72.7	2.942	132.6141935	8.099
15	13-08-2010	622437.08	0	72.17	2.923	132.63	8.099
15	20-08-2010	637090.44	0	71.1	2.913	132.6616129	8.099
15	27-08-2010	649791.15	0	65.07	2.885	132.6932258	8.099
15	03-09-2010	638647.21	0	73.17	2.86	132.7248387	8.099
15	17-09-2010	583210.87	0	57.94	2.846	132.7670667	8.099
15	24-09-2010	548542.47	0	62.53	2.837	132.7619333	8.099
19	02-07-2010	1549018.68	0	66.25	2.958	132.5218667	8.099
19	09-07-2010	1577541.24	0	78.22	2.94	132.5461333	8.099
19	16-07-2010	1412157.02	0	74.6	2.933	132.5667742	8.099
16	02-07-2010	1373043.71	0	74.69	3.034	133.6316096	8.099

Dashboard:



An interactive Walmart Weekly Sales Dashboard was successfully developed using Power BI and the Walmart Weekly Sales Dataset. The dashboard provides a comprehensive overview of sales performance through KPI cards and interactive visualizations, enabling users to analyze weekly sales trends, compare store-wise performance, evaluate holiday and non-holiday sales, identify top-performing stores, and examine the relationship between temperature and weekly sales. Interactive slicers for Date, Store, Holiday Flag, Temperature, Fuel Price, CPI, and Unemployment allow dynamic filtering and improve the overall user experience. The dashboard transforms raw sales data into meaningful business insights, making it easier for management to monitor performance, identify sales patterns, and support data-driven decision-making through an intuitive and visually appealing reporting interface.

Conclusion

The **Walmart Weekly Sales Dashboard** was successfully designed and developed using **Power BI** to provide an interactive and comprehensive analysis of weekly sales data. The dashboard integrates **KPI cards, interactive slicers, line charts, bar charts, donut charts, and scatter charts** to present meaningful business insights in a clear and user-friendly manner. It enables users to monitor weekly sales trends, compare store performance, evaluate holiday and non-holiday sales, and analyze the relationship between temperature and sales through dynamic filtering and visualization. Overall, the dashboard effectively transforms raw sales data into actionable insights, supporting data-driven decision-making and demonstrating the practical application of Power BI in business intelligence and sales performance analysis.

Learning Outcomes

After completing this experiment, I learned:

- How to import, clean, and prepare business data in Power BI for analysis.
- How to transform raw sales data and configure appropriate data types for effective reporting.
- How to create DAX measures to calculate Key Performance Indicators (KPIs) such as Total Sales, Average Weekly Sales, Total Stores, and Holiday Weeks.
- How to design an interactive business dashboard using KPI Cards, Line Charts, Clustered Bar Charts, Donut Charts, and Scatter Charts.
- How to create and configure interactive slicers for Date, Store, Holiday Flag, Temperature, Fuel Price, CPI, and Unemployment to enable dynamic data filtering.
- How to apply professional dashboard design principles, including layout planning, formatting, color themes, alignment, and visual consistency.
- How to implement interactive features such as navigation buttons, bookmarks, cross-filtering, and a Reset Filters button to improve user experience.
- How to analyze weekly sales trends, compare store performance, evaluate holiday and non-holiday sales, and study the relationship between temperature and sales using Power BI visualizations.
- How to transform raw business data into meaningful insights that support data-driven decision-making and effective business reporting.

GitHub Repository:

<https://github.com/deepakMudiraj06/Power-BI/tree/main/Assignment%205>